

Skill-02 Task 01 - Main Shell Loop

Objective

To create the main interactive loop of Shellforge by displaying a prompt, reading user input, handling exit conditions, designing control flow, and testing the interactive loop.

Features

- Display shell prompt
- Read user input
- Support multi-character commands
- Handle empty input
- Detect exit command
- Terminate shell using exit condition

Source File

main_loop.c

Compilation

```
gcc main_loop.c -o main_loop
```

Execution

```
./main_loop
```

Sample Output

```
Shellforge> hello Command entered: hello

Shellforge> ls Command entered: ls

Shellforge> test Command entered: test

Shellforge> exit Exiting Shellforge...
```

Result

The interactive shell loop was successfully implemented and tested.

```
#include <stdio.h>

#include <string.h>

int main()
{
    char input[100];

    while (1)
    {
        printf("Shellforge> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            break;
        }

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
        {
            printf("Exiting Shellforge...\n");
            break;
        }

        if (strlen(input) == 0)
```

```
{  
    continue;  
}  
  
printf("Command entered: %s\n", input);  
}  
  
return 0;  
}
```

Output-

Shellforge> hello

Command entered: hello

Shellforge> ls

Command entered: ls

Shellforge> test

Command entered: test

Shellforge> exit

Exiting Shellforge...